Data Warehouse Design Justification

The DIM_COUNTRY, DIM_TIME, and DIM_SECTOR dimensions use integer surrogate keys (country_key, time_key, and sector_key) that are independent of source codes and descriptions. Natural keys are retained as identifying attributes where applicable; for example, iso3_code identifies a country and year identifies a year.

The three fact tables represent different business processes and grains:

- FACT_ECONOMY: one row per country and year. It stores economic indicators such as GDP, GDP per capita, and international trade.
- FACT_SOCIETY: one row per country and year. It stores social and demographic indicators such as population, GNI per capita, life expectancy, and urbanization.
- FACT_SECTOR: one row per country, year, and sector. It stores measures of economic structure and employment by sector.

FACT_ECONOMY and FACT_SOCIETY share a country-year grain, but they describe different subject areas and contain different sets of measures. FACT_SECTOR has a more detailed country-year-sector grain. Keeping the tables separate avoids mixing processes with different grains, duplicating measures when joining rows by sector, and creating columns that have no meaning for some processes. The UNIQUE constraints enforce the intended grains.

DIM_SECTOR is connected only to FACT_SECTOR because this is the only fact table whose measures are broken down by sector. GDP, population, and life expectancy at the country-year level are not associated with a sector in this model. Connecting those fact tables to DIM_SECTOR would add no information and could imply a breakdown that does not exist.

Historical Events Integration

The chosen approach is a reference/bridge table, EVENTS, connected to DIM_COUNTRY and DIM_TIME through country_key and time_key. Each row represents an event and stores its description, category, and qualitative economic impact. This approach allows multiple events to be associated with the same country and year without assigning quantitative measures to events or changing the grain of the fact tables. event_key is a surrogate key for the EVENTS table; country_key and time_key reference the surrogate keys in the dimensions. EVENTS is not a fourth fact table of economic indicators, and no DIM_EVENT was created because events are used as descriptive context rather than as an analytical dimension for the measures in this design.

The accompanying ER diagram represents this model and shows the three fact tables, the three dimensions, the EVENTS table, and their foreign-key relationships.